5 YEARS IN .NET

INTERVIEW PREP

Interview Questions for .NET Developers

25 questions I've been asked — and asked others — across banking, insurance & healthcare domains.

CORE .NET & C#

1. What is the difference between IEnumerable, IQueryable, and IList?
2. Explain async/await. What happens if you use async void?
3. What are Transient, Scoped, and Singleton lifetimes in DI?
4. How does garbage collection work in .NET? What are Gen 0, 1, 2?
7. What is middleware in ASP.NET Core? How does the pipeline work?
8. How does Entity Framework Core handle lazy vs eager loading?
9. What is the Repository pattern and when would you NOT use it?
10. How do you prevent N+1 query problems in EF Core?

5 What is the difference between value types and reference types?

6 Explain the SOLID principles with a real example in C#.

11 What is the difference between JWT and session-based auth?

12 Explain how OAuth2 and OpenID Connect differ.

ARCHITECTURE & SYSTEM DESIGN

13 How would you design a rate limiter for a .NET API?

14 When would you choose REST over gRPC for microservices?

15 What is the Outbox pattern and why does it solve distributed transaction problems?

16 How do you implement caching in ASP.NET Core? Redis vs memory?

17 Explain the Circuit Breaker pattern. When should it be applied?

18 How would you handle database migrations in a zero-downtime deployment?

19 What is CQRS and when does it add complexity without value?

20 How do you secure sensitive config values in Azure App Service?

21 Explain your approach to API versioning in ASP.NET Core.

22 How would you design an audit log system for a banking app?

SQL & PERFORMANCE

23 What is the difference between clustered and non-clustered indexes?

24 How do you identify and fix a slow SQL query in production?

25 Explain database isolation levels and when you'd change the default.

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Full Stack .NET Developer | ASP.NET Core · Angular · Azure

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